NIH ChestX-ray14

Dataset Card & Descriptive Profile

Documentation for a binary chest X-ray classification research project, covering dataset identity, metadata, label construction, descriptive statistics, usage conditions, limitations, and responsible use.

IMAGES	UNIQUE PATIENTS	IMAGE FORMAT	ACCESS DATE
112,120	30,805	PNG	27 Oct 2026

Official dataset source

nihcc.app.box.com/v/ChestXray-NIHCC

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Scope note. This document combines dataset documentation with descriptive statistics supplied for the project. It is not a clinical validation report, and descriptive associations should not be interpreted as proof of model bias or performance.

Image dimensions note

Dimensions vary in the working metadata; images should not be described as uniformly 1024 × 1024 pixels. Width: mean 2,646.08 (range 1,143–3,827); height: mean 2,486.44 (range 966–4,715). These values do not establish later resizing or cropping.

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Project label convention

Normal (0) is assigned when `finding_labels` is exactly `No Finding`. **Abnormal (1)** is assigned when one or more disease findings are present. This is a project-specific binary transformation of the original labels.

1. Dataset Identity and Source

Dataset name: NIH ChestX-ray14

Provider: National Institutes of Health (NIH) Clinical Center

Official source: NIH ChestX-ray dataset download page

Access date recorded for this project: 27 October 2026

The dataset contains 112,120 frontal-view chest radiographs associated with 30,805 unique patients. The images are provided in PNG format, and the working metadata records variable image dimensions. The dataset package also includes a CSV file with image-level metadata and finding labels. The approximate package size can vary depending on packaging or extraction.

Primary metadata file: `Data_Entry_2017_v2020.csv`. Other files and folders may be present in the official download package; this card focuses on the fields and statistics used in the current project.

Dataset summary

Property	Project value
Images / image records	112,120
Unique patients	30,805
Image format	PNG
Reported image dimensions	Variable; see note on cover
Metadata source	Data_Entry_2017_v2020.csv
Image view	Frontal views; AP and PA are represented

Source and attribution requirements are summarized in Section 8.

2. Dataset Structure and Metadata

The CSV file contains image-level records. In this project, the columns have been renamed to make their roles clearer. The table below documents the core fields used for classification and reliability analysis.

Field	Meaning and project use
`image_name`	Image file name or identifier used to locate the image.
`finding_labels`	Original finding label string: No Finding, one disease label, or multiple disease labels separated by a delimiter.
`follow_up`	Follow-up number associated with the image record; it is not itself a time interval.
`patient_id`	Patient identifier. Used to keep all images from a patient in a single data split.
`patient_age`	Patient age as recorded in the dataset.
`patient_sex`	Sex value as recorded in the dataset.
`view_position`	Image acquisition view, such as AP or PA.
`image_width, image_height`	Recorded image dimensions.
`pixel_spacing_x, pixel_spacing_y`	Pixel-spacing information in the two image directions, when available in the working metadata.

Why metadata matters. Patient identifiers are essential for leakage-resistant splitting. Age, sex, and view position support subgroup-level descriptive analysis. Metadata should be treated as recorded information and not assumed to be error-free.

3. Label Definition and Construction

3.1 Original finding labels

The original `finding_labels` field may contain `No Finding`, a single disease label, or multiple disease labels. These labels were automatically extracted from associated radiology reports using natural-language-processing methods rather than being independently assigned by a radiologist reviewing every image for this dataset release.

3.2 Binary labels used in this project

Project label	Definition	Rule applied
Normal (0)	No finding label recorded	`finding_labels == 'No Finding'`
Abnormal (1)	At least one disease finding recorded	Any label other than `No Finding`, including multi-label strings

> **Important interpretation.** The binary target is a project-defined simplification of the original multi-label task. “Normal” means that the dataset label is `No Finding`; it should not be interpreted as a guarantee that the image is clinically normal. Similarly, “Abnormal” means that one or more findings were recorded, not that every finding is confirmed.

3.3 Label reliability caveat

For the original dataset, the authors used a natural-language-processing (NLP) labeler to process the text of radiology reports associated with the X-rays and automatically extract disease findings. The resulting labels were therefore generated from report text, rather than being independently confirmed by a radiologist for every image. The extraction process can make mistakes or miss findings, and a report label may not perfectly match what is visible in the image. These labels should be treated as automatically extracted, weak labels—not perfect ground truth. Label errors may affect both model training and the evaluation of predictions.

4. Dataset Profile and Descriptive Statistics

4.1 Binary label distribution

The current project mapping yields the following distribution across all 112,120 image records:

Binary class	Label	Image count	Share
Normal	0	60,361	53.84%
Abnormal	1	51,759	46.16%
Total	—	**112,120**	**100%**

The difference between the two classes is 8,602 images, with an approximate Normal-to-Abnormal ratio of 1.17:1. This is a moderate imbalance in the binary labels. The distribution should still be considered when selecting metrics and interpreting model results.

4.2 Original disease-label distribution

The original dataset includes 14 disease categories plus `No Finding`. Images can carry multiple disease labels, so disease-label percentages are not mutually exclusive and do not sum to 100%.

Original label	Image count	Share of images
No Finding	60,361	53.84%
Infiltration	19,894	17.74%
Effusion	13,317	11.88%
Atelectasis	11,559	10.31%
Nodule	6,331	5.65%
Mass	5,782	5.16%
Pneumothorax	5,302	4.73%
Consolidation	4,667	4.16%
Pleural_Thickening	3,385	3.02%
Cardiomegaly	2,776	2.48%
Emphysema	2,516	2.24%
Edema	2,303	2.05%
Fibrosis	1,686	1.50%
Pneumonia	1,431	1.28%
Hernia	227	0.20%

Observed pattern: Infiltration is the most frequent disease label, while Hernia is the rarest. Disease-specific counts should not be added together to obtain a total number of images because multi-label images contribute to more than one category.

5. Acquisition and Demographic Characteristics

5.1 AP and PA view positions

View	Image count	Share
PA	67,310	60.0%
AP	44,810	40.0%

AP (anteroposterior) means the X-ray beam passes from the front of the chest toward the back; PA (posteroanterior) means it passes from the back toward the front. PA images are commonly taken with the patient upright, while AP images are often used for portable or bedside exams. In this dataset, about 53% of AP images are labeled Abnormal. For PA images, 58.39% are labeled Normal and 41.61% Abnormal. These differences may reflect patient and acquisition context; the percentages alone do not prove model bias or different model performance.

5.2 Sex distribution by binary label

Recorded sex	Normal	Abnormal
F	54.20%	45.80%
M	53.56%	46.44%

The reported binary-label percentages are similar across the recorded F and M groups. The difference in Abnormal share is approximately 0.64 percentage points. This descriptive comparison alone does not establish equal model performance across groups; that requires evaluating model predictions with suitable subgroup metrics.

5.3 Patient age distribution

Statistic	Normal (0)	Abnormal (1)
Mean age	45.48	47.96
Standard deviation	16.53	16.59
Median age	47	50
Interquartile range (Q1–Q3)	33–58	36–60
Minimum–maximum	0–95	0–94

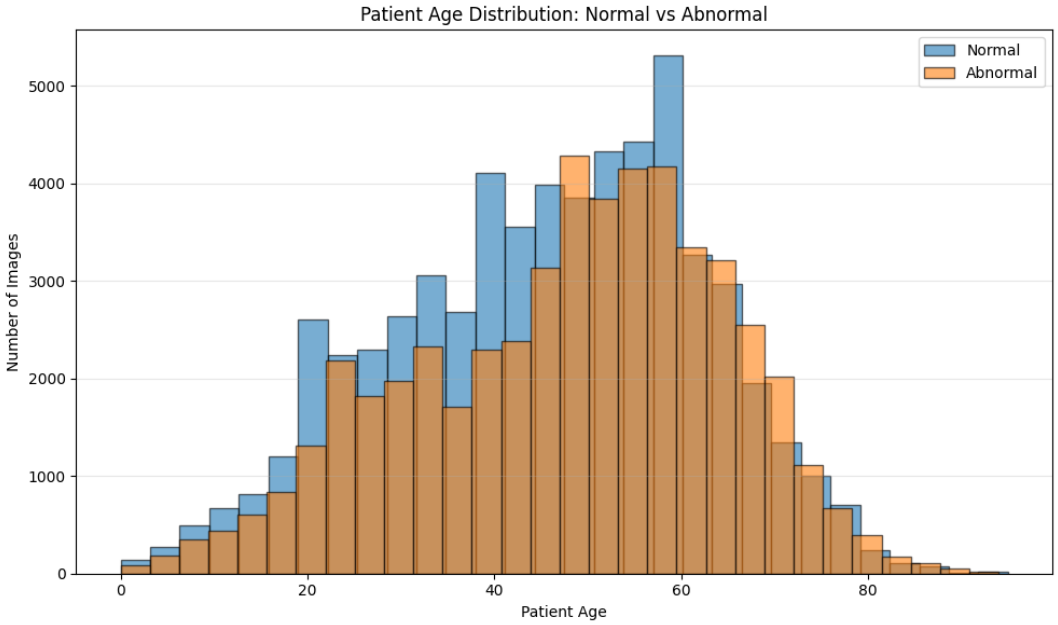


Figure 1. Image-count distribution by patient age for the Normal and Abnormal binary classes.

The distributions overlap substantially, while the Abnormal group has a slightly higher mean and median age. The plot counts image records by age, not unique patients. Fourteen patient IDs were flagged in the supplied analysis as having recorded age 0; this should be reviewed as a data-quality question rather than automatically corrected.

6. Patient-Level Repetition and Data Leakage

The dataset contains multiple images for many patients. The supplied patient-level summary is shown below.

Measure	Value
Unique patients	30,805
Mean images per patient	3.64
Median images per patient	1
Minimum–maximum images per patient	1–184
Patients with more than one image	13,302 (43.18%)
Patients with one image	17,503 (56.82%)

A random image-level split can place images from the same patient in both training and evaluation sets. Because images from one patient may share anatomy, acquisition characteristics, or repeated-examination context, this can lead to leakage and an overly optimistic estimate of generalization.

> **Project recommendation:** split data by `patient_id`, not by image. Every image associated with a patient should belong to exactly one of the training, validation, or test partitions. Verify this condition after splitting.

Follow-up number

Statistic	Value
Mean	8.57
Median	3
Minimum–maximum	0–183

The mean exceeding the median suggests a right-skewed distribution, with some larger follow-up numbers. These values describe the follow-up-number column; they do not measure the elapsed time between examinations.

7. Data Quality Notes and Known Limitations

Automatically extracted labels

Labels are derived automatically from radiology reports and may contain noise, omissions, or mismatches. They should not be treated as independently verified ground truth for every image.

Binary-label simplification

The project collapses the original multi-label problem into Normal versus Abnormal. This removes disease-specific information and makes the meaning of Normal dependent on the original No Finding label.

Repeated examinations

Multiple images per patient make patient-level partitioning essential to reduce data leakage.

AP/PA distribution differences

AP and PA images are represented in different proportions and have different observed binary-label compositions. View-stratified reporting may help characterize model behavior.

Demographic representation

Age and recorded sex vary across records. The supplied summary is descriptive and is not sufficient to establish demographic fairness or equal error rates.

Rare disease categories

The original disease-label counts vary substantially; Hernia is much rarer than Infiltration. Disease-specific analyses may be less stable for low-count categories.

Age values recorded as zero

The supplied analysis flagged 14 patient IDs with age recorded as 0. Their meaning should be checked against the project's data-handling policy or discussed with the supervisor before any correction or exclusion.

Image quality and acquisition confounders

This summary does not include a complete image-quality audit, duplicate-image analysis, or image-embedding analysis. Claims about these factors should be made only after those analyses are completed.

Missing values

The project analysis reports no missing values in the working metadata table. This does not guarantee that every field is clinically valid or that all potential data-quality issues have been resolved.

> **Interpretation rule:** descriptive differences identify questions to investigate; they do not, by themselves, demonstrate causation, model bias, or clinical usefulness.

8. Licensing, Attribution, and Citation

The usage information collected for this project indicates that NIH ChestX-ray14 may be used without restrictions, subject to attribution requirements. The documentation reviewed for this card does not identify a separate standardized license such as CC BY or CC0. Users should consult the official NIH package documentation and FAQ for the terms applicable to their use.

Attribution steps

• Link to the original NIH dataset download page: https://nihcc.app.box.com/v/ChestXray-NIHCC.

• Acknowledge the NIH Clinical Center as the data source.

• Cite the original CVPR 2017 publication by Wang and colleagues.

Suggested citation

Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. (2017). *ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases.* Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 3462–3471. https://doi.org/10.1109/CVPR.2017.369.

> **Attribution statement for project materials:** “Data source: NIH Clinical Center, NIH ChestX-ray14. Please refer to the official dataset page and Wang et al. (2017) for dataset documentation and the original publication.”

9. Ethics and Responsible Use

The NIH Clinical Center states that the released chest X-ray images were anonymized before public release. The project will use the dataset responsibly for research and educational purposes, respect patient privacy, and avoid any attempt to re-identify individuals.

This project is a research prototype intended to study chest X-ray classification and model reliability. It is not a clinically validated diagnostic tool and must not be used as a substitute for professional radiological interpretation or to make clinical decisions. Model predictions may contain errors and may be affected by label noise, class imbalance, patient-level repetition, and differences in image acquisition. Results should be interpreted within these limitations.

Public availability and de-identification do not automatically settle every institutional or regulatory requirement. Any future publication, deployment, or use beyond educational research should follow the applicable institutional policies and relevant requirements.

Responsible-use checklist

- ■ Do not attempt to identify or contact individuals represented in the dataset.
- ■ Use patient-level data splitting and report the split procedure.
- ■ Report class-specific and subgroup metrics rather than relying only on overall accuracy.
- ■ Describe labels as automatically extracted and acknowledge possible label noise.
- ■ Do not present model outputs as medical diagnoses or treatment recommendations.
- ■ Document the dataset source and include the required attribution and citation.

10. References

1. NIH Clinical Center. NIH Chest X-ray dataset download page.

https://nihcc.app.box.com/v/ChestXray-NIHCC

2. Google Cloud. NIH Chest X-ray public dataset documentation and usage information.

https://docs.cloud.google.com/healthcare-api/docs/resources/public-datasets/nih-chest

3. Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. (2017). ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. CVPR.

https://doi.org/10.1109/CVPR.2017.369

4. NIH Intramural Research Program. NIH Clinical Center provides one of the largest publicly available chest X-ray datasets (anonymization and release context).

https://irp.nih.gov/news-and-events/in-the-news/nih-clinical-center-provides-one-of-the-largest-publicly-available-chest-x

5. RSNA Atlas. NIH ChestX-ray14 dataset card (dataset description and label provenance context).

https://atlas.rsna.org/cards/66962171-1dcb-431d-89b0-08a561f1c8ed

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Document note: Counts and descriptive statistics in this card are based on the project-provided analysis. Before publication, verify the exact working CSV schema, dataset version, and current terms against the official NIH files.